GATE-KD: Aligning Teacher Targets to the Student Space

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Abstract

1 Method

Let $f_T : \mathcal{X} \to \mathbb{R}^{d_T}$ be a frozen teacher and $f_S(\cdot;\theta) : \mathcal{X} \to \mathbb{R}^{d_S}$ a student with $d_T > d_S$. We run the teacher once over an unlabeled corpus $\{x_i\}_{i=1}^N$ and cache its normalized final embeddings

$$\mathbf{t}_i = \mathrm{norm}(f_T(x_i)), \qquad \mathrm{norm}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}. \tag{1}$$

The corresponding deployed student embedding is $\mathbf{z}_i(\theta) = \mathrm{norm}(f_S(x_i;\theta))$. Both are represented as row vectors.

GATE-KD first selects a fixed student-width subspace with P_{PCA}, then orients it toward the student with a closed-form $R^{(e)}$. Neither matrix receives gradients; only θ is learned. The construction is summarized in Algorithm 1.

1.1 Student-Aligned Target Construction

Selecting a student-width subspace. Let $\mu = N^{-1}\sum_i \mathbf{t}_i$. We fit the leading d_S principal directions of the centered teacher cache and collect them as $P_{\mathrm{PCA}} \in \mathbb{R}^{d_T \times d_S}$, where $P_{\mathrm{PCA}}^\top P_{\mathrm{PCA}} = I_{d_S}$. We then construct student-width targets

$$\mathbf{y}_i = \mathrm{norm}(\mathbf{t}_i P_{\mathrm{PCA}}) \in \mathbb{S}^{d_S-1}. \tag{2}$$

Centering is used only to fit the directions; we apply neither mean subtraction nor whitening. PCA is a deterministic default, not a requirement: the next step applies to any fixed student-width target matrix (Appendix **??**).

Choosing a student-relative orientation. Dimensional reduction determines which teacher variation is retained but not how the resulting coordinates should be oriented. If $Y \in \mathbb{R}^{N\times d_S}$ collects the rows $\mathbf{y}_i$, then

$$[Y]_{O(d_S)} = \{YR : R \in O(d_S)\} \tag{3}$$

contains equally valid orientations of the reduced targets. Every member preserves the same pairwise geometry because $(YR)(YR)^\top = YY^\top$, yet a pointwise endpoint loss assigns different gradients to different members. GATE-KD therefore selects one orientation relative to the student rather than learning an additional map.

Let $\mathcal{C}$ be a fixed gauge-fit set, equal to the complete distillation corpus in our experiments. At the start of epoch e, we encode $\mathcal{C}$ with the student in evaluation mode and collect the embeddings in $Z_{\mathcal{C}}^{(e)}$. For the matching rows $Y_{\mathcal{C}}$, we solve the orthogonal Procrustes problem (Schönemann, 1966):

$$\begin{aligned} U\Sigma V^\top &= \mathrm{svd}\left(Y_{\mathcal{C}}^\top Z_{\mathcal{C}}^{(e)}\right), \\ R^{(e)} &= UV^\top \in \operatorname*{arg\,min}_{R\in O(d_S)} \left\| Y_{\mathcal{C}} R - Z_{\mathcal{C}}^{(e)} \right\|_F^2. \end{aligned} \tag{4}$$

The target for sentence x_i during epoch e is

$$\boldsymbol{\tau}_i^{(e)} = \mathbf{y}_i R^{(e)}. \tag{5}$$

All minibatches in epoch e share this target, and $R^{(e)}$ is constant during student updates. It acts only on cached teacher targets and is absent from the deployed student. In particular, $R^{(0)}$ is an interface for one pretrained student checkpoint, not a transferable teacher coordinate frame.

A corpus-global construction is more restrictive than minimizing over orientation independently within each minibatch. For a *fixed* pair of representations (Y, Z), this distinction in attainable objective values can be stated exactly.

Proposition 1 (Batchwise Procrustes relaxation). *For any partition $\{\mathcal{B}_k\}_{k=1}^{K}$ of the corpus, let $C_k = Y_{\mathcal{B}_k}^\top Z_{\mathcal{B}_k}$. The locally and globally rotation-minimized mean cosine losses satisfy*

$$\mathcal{L}_{\text{local}} = 1 - \frac{1}{N}\sum_k \|C_k\|_* \;\le\; 1 - \frac{1}{N}\left\|\sum_k C_k\right\|_* = \mathcal{L}_{\text{global}}, \tag{6}$$

where $\|\cdot\|_*$ is the nuclear norm. Equality holds if and only if one orthogonal rotation is simultaneously optimal for every minibatch.

Proof. For any matrix C, orthogonal Procrustes gives $\|C\|_* = \max_{R\in O(d_S)} \operatorname{tr}(R^\top C)$. Maximizing the sum with one R cannot exceed maximizing each term with its own R_k, which yields Equation 6. Equality requires one R to maximize every term, and this condition is sufficient. □

This is a statement about fixed (Y, Z), not training or generalization. A local zero loss may use different R_k across batches, preserving each within-batch Gram matrix while changing cross-batch similarities through $R_a R_b^\top$. Table **??** changes additional factors and is not a causal test of this freedom.

A learned map can leave the deployed representation non-identifiable. For any invertible A, setting $\mathbf{z}_i = \operatorname{norm}(\mathbf{y}_i A)$ and learning $W = A^{-1}$ gives $\operatorname{norm}(\mathbf{z}_i W) = \mathbf{y}_i$, hence zero mapped cosine loss, while $ZZ^\top$ generally differs from $YY^\top$ (Bhattarai et al., 2025). GATE-KD instead restricts the target-side transformation to $O(d_S)$, which preserves $YY^\top$, and applies the loss directly to the embedding used at inference. This identifies the geometric freedom excluded by our interface, not a claim that projectors are generally ineffective.

1.2 Optional Native-Space H_0 Auxiliary

The gauge-fixed endpoint target provides the primary sample-level supervision. As an optional coordinate- and dimension-independent signal, we also match a coarse summary of connectivity scales in the original teacher and student spaces. For a minibatch $\mathcal{B}$ of size B, we use chord distance

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u}, \mathbf{v}\rangle}. \tag{7}$$

The $B-1$ finite death times of degree-zero Vietoris–Rips persistence equal the edge weights of the point cloud's minimum spanning tree (Hofer et al., 2020). Let $q^S_{(1)} \le \cdots \le q^S_{(B-1)}$ and $q^T_{(1)} \le \cdots \le q^T_{(B-1)}$ be the sorted death times computed from the student embeddings $\{\mathbf{z}_i : i \in \mathcal{B}\}$ and the original teacher embeddings $\{\mathbf{t}_i : i \in \mathcal{B}\}$. We define

$$\mathcal{L}_{H_0} = \frac{1}{B-1}\sum_{j=1}^{B-1}\left(q^S_{(j)} - q^T_{(j)}\right)^2. \tag{8}$$

The teacher summary uses $\mathbf{t}_i$, so dimensions and orientation may differ. MST selection and sorting are detached; gradients flow through selected student edges. We use $B = 128$, FP32 distances, $\varepsilon = 10^{-7}$, and SciPy's upper-triangle input order to break ties. The loss matches only the distribution of connectivity scales; Algorithm **??** gives details.

1.3 Student Optimization

For the same minibatch $\mathcal{B}$, the endpoint loss is

$$\mathcal{L}^{(e)}_{\text{end}} = \frac{1}{B}\sum_{i\in\mathcal{B}}\left(1 - \left\langle \mathbf{z}_i, \boldsymbol{\tau}_i^{(e)} \right\rangle\right). \tag{9}$$

The complete objective is

$$\mathcal{L}^{(e)} = \mathcal{L}_{\text{end}}^{(e)} + \lambda_{\text{topo}} \mathcal{L}_{H_0}. \tag{10}$$

The headline results use $\lambda_{\text{topo}} = 0.5$; the endpoint-only control sets it to zero. Our main recipe refits the corpus-global orientation after each non-final epoch; the fixed-$R^{(0)}$ schedule is evaluated as a control. A refresh requires one student pass over $\mathcal{C}$ and an SVD of a $d_S \times d_S$ matrix, with no backpropagation through either operation. Teacher embeddings and the PCA subspace are computed once, so ordinary minibatch updates require no online teacher pass. At inference, GATE-KD uses only the student's normalized final embedding and introduces no additional parameters or computation.

Algorithm 1 GATE-KD training with a non-trainable target-side interface

Require: Frozen teacher f_T; initialized student $f_S(\cdot;\theta)$; corpus $\mathcal{C}$; epochs E; refresh epochs $\mathcal{E}_R \subseteq \{0,\ldots,E-1\}$ with $0 \in \mathcal{E}_R$; $\lambda_{\text{topo}} \geq 0$.

Cache $\mathbf{t}_i \leftarrow \operatorname{norm}(f_T(x_i))$ for every $x_i \in \mathcal{C}$.
Fit P_{PCA} once and construct $\mathbf{y}_i \leftarrow \operatorname{norm}(\mathbf{t}_i P_{\text{PCA}})$.
for $e = 0,\ldots,E-1$ **do**
    if $e \in \mathcal{E}_R$ **then**
        In evaluation mode and without gradients, encode $\mathcal{C}$ to obtain $Z_{\mathcal{C}}^{(e)}$.
        Compute $U\Sigma V^\top \leftarrow \operatorname{svd}(Y_{\mathcal{C}}^\top Z_{\mathcal{C}}^{(e)})$ and set $R^{(e)} \leftarrow UV^\top$.
    else
        Set $R^{(e)} \leftarrow R^{(e-1)}$.
    end if
    for each minibatch $\mathcal{B} \subset \mathcal{C}$ **do**
        Compute $\mathbf{z}_i(\theta)$ and $\boldsymbol{\tau}_i^{(e)} \leftarrow \mathbf{y}_i R^{(e)}$ for $i \in \mathcal{B}$.
        Compute $\mathcal{L}_{\text{end}}^{(e)}$ using Equation 9; set $\mathcal{L}_{H_0} \leftarrow 0$ when $\lambda_{\text{topo}} = 0$, and otherwise use Equation 8.
        Update only θ using $\nabla_\theta(\mathcal{L}_{\text{end}}^{(e)} + \lambda_{\text{topo}} \mathcal{L}_{H_0})$.
    end for
end for
return $f_S(\cdot;\theta)$.

References

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